Taizhen Cheung

U.S. Citizen | Brooklyn, NY

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Education

Hunter College, CUNY **Expected May 2027**

B.A., Double Major in Computer Science and Mathematics

Coursework: Data Structures, Algorithms, Software Analysis & Design, Linear Algebra, Numerical Methods

Technical Skills

Languages: Python, C++, TypeScript, JavaScript, SQL, HTML/CSS

Frameworks: React, Next.js, Preact, Node.js, Express.js, FastAPI, PyTorch, NumPy, Pandas

Data & Tools: PostgreSQL, Supabase, pgvector, SQLite, Docker, Git, GitHub Actions, Linux, pytest, Playwright

Selected Projects & Research

Onward | AI Storytelling Application | Live app **May 2026 - Present**

Next.js, TypeScript, Supabase, PostgreSQL, LLMs, RAG

- Built and deployed an app to help users facing personal setbacks find relatable historical experiences; matched disclosures to 50 documented life episodes and delivered seven-part stories through server-sent events.
- Implemented keyword search with LLM reranking and a vector-search challenger; evaluated story relevance on 104 synthetic cases, with keyword reranking reaching 96.0% top-1 accuracy across 202 matchable trials.
- Protected private stories with anonymous authentication, server-side ownership checks, and PostgreSQL rate limits; implemented automatic guest-account and story deletion after about six hours of inactivity.

resufill | Resume Tailoring Tool | Code **Aug 2026**

Python, FastAPI, Pydantic, Preact, TypeScript, Playwright

- To tailor applications without inventing experience, built a CLI and web UI that trace generated claims to career records, reject unsupported statements, and report missing job qualifications as gaps.
- Implemented source-by-source audits and Chromium PDF export with text-extraction checks; made generation fail if contact details, headings, or bullets are lost during export, backed by automated regression tests.

When Directional Accuracy Lies | Independent Research | Code **Mar 2026 - Jul 2026**

Python, PyTorch, TimesFM, LoRA | Sole-author preprint: arXiv:2607.12248

- Built a leakage-controlled, walk-forward benchmark to test an apparent 80% directional accuracy; found no consistent advantage over always-up predictions across S&P 500 and NASDAQ-100 stocks.
- Released a sole-author arXiv preprint, checksum-verified datasets, committed results, and 55 CPU tests covering splits, metrics, and baselines, enabling reproduction of the evaluation tables and figures.

xiaofinance | Investment Discussion Dashboard | Code **Jul 2026**

- Built a FastAPI/SQLite dashboard to organize 24-hour Xiaohongshu investment discussions; used simhash to collapse reposts and validated summary quotations against source text to make sentiment reports traceable.

Experience

Handshake | AI Trainer - Coding & Image Processing **Nov 2025 - Aug 2026**

- Evaluated multi-turn coding responses for correctness and instruction following; produced comparative rankings and evidence-based rationales for AI coding-assistant feedback datasets.
- Used Python tooling and structured annotation to review multimodal training examples; identified labeling inconsistencies and bias risks, documenting quality findings against project requirements.

TIER Lab, Hunter College | Research Intern **Jun 2025 - Sep 2025**

- Benchmarked 1,000+ prompts across Gemini 2.0 Flash, DeepSeek R1, and GPT-4o using reproducible Python datasets; compared reasoning consistency, safety, latency, and bias to inform model evaluation.
- Designed bias audits for LLM-integrated robotics and human-robot interaction; analyzed latency and bias trade-offs and delivered mitigation recommendations to research stakeholders.